Additional file 6: Reproducibility details

These records describe retained computational evidence. The complete inputs are provided through the v0.2.0 study archive identified in the main article. Full stored-map replay is distinct from verification of exported scalar results. Provider output terms and modification notices accompany the archive.

The within-site analyses and their numerical record are archived in the existing AlphaGenome reproduction package [1]. Its source code, curation records and supplementary tables give the complete empirical methods. An extraction manifest records selected machine-artifact values and hashes; extraction does not constitute independent regeneration of all earlier analyses. A separate targeted cache check reproduces the three reported AVI stratum counts and mean statistics and records their tie frequencies; it does not rerun their permutation tests. The schema probe and mathematical-example verifier accompany the present analysis. The verifier checks interval translation, scale normalization, paired diagram-alignment bounds, nonmonotonicity under vertex deletion, the distinction between bottleneck and Betti-curve distances, the failure of treating an arbitrary contact transform as a metric, tie-aware conditional nulls and rank invariance. No result from these finite examples is labeled a genomic observation.

The completed spatial reproduction bundle includes all raw per-call reference/alternate maps, assembled site arrays, coordinates, track metadata, input hashes, the locally frozen protocol, exact-matching settings, full diagrams, all sixteen conditional tests, numerical diagnostics and acquisition audits. The AlphaGenome client was version 0.9.0; its non-generated Python source matched the recorded vendored commit. ALL_FOLDS is a model alias; an immutable server weight revision was not exposed. The strengthened mathematical verifier passed eight check groups, including 100 randomized quotient comparisons. Independent inference checks include exhaustive finite-label distributions. These finite checks do not by themselves constitute a Lean formalization, a clean-environment regeneration of the retained audit, or biological validation.

The anchor extension additionally contains the frozen replication cohort and protocol, all 84 sites' raw predictions, native diagrams, 252 endpoint comparisons, 21 direct midpoint checks, scalar reconstruction baselines, known-answer controls and independent derived-count checks. Exact rational finite checks exercise the local slope and endpoint-envelope arguments; independent Hera checks include turning and flat controls. The prospective engine retains failures and source hashes, and did not exclude any planned site. The numerical comparison tolerances and their limits accompany the outputs. Additional file 2 supplies the full proofs and counterexamples, and the integration map supplied with the review package records how this replacement incorporates the earlier contribution.

A subsequent validation pass regenerated all 204 spatial sites from cached maps in a fresh environment, including full diagrams, endpoint distances and the fixed midpoint checks. All 6,324 per-site comparisons passed the declared tolerances; 18 centered-distance vectors differed only at floating-point rounding scale. All sixteen conditional tests were independently recertified from the new outputs. The six retained historical output files regenerated byte-for-byte using their retained original random seeds. This checks execution reproducibility, not Monte Carlo precision. Linkage to each output is recorded in the validation report. The portable reproduction package pins its dependencies, inputs and expected outputs and preserves the failed setup and postprocessing attempts alongside their repairs. This reproduction checks computations on the retained inputs; it does not establish model accuracy or immutable server weights.

The cross-track extension retains its pre-acquisition protocol, all 252 two-track calls, the independently

checked GM12878 drift control, the native-prediction lock, per-site diagrams and centered validations, and a full reference-engine replay using the original persistent-homology implementation. A separate computational audit recomputed Hausdorff witnesses by brute force and reconstructed rankings and rational selected-allele credits from per-site files. The only execution correction was an initial import failure in the acquisition environment; the unchanged analysis ran in the existing fresh validation environment. Failed import logs and actual dependency versions are preserved. The portability validation report supplied with the review package distinguishes the new checks from the preceding full harness and its remaining human and provenance requirements.

Targeted Julia persistence check

A subsequent targeted check used Ripserer 0.17.0 [2] under Julia 1.13.0 to reconstruct full H1 diagrams from saved maps. The sample combined the previously frozen smallest-hash site in each of the seven replication genes (indices 7, 21, 34, 44, 59, 70 and 75) with every site containing a known IMR-90 nonsaturation exception (48, 51 and 59). The union comprised nine sites on both GM12878 and IMR-90 tracks. Reference and all three alternate maps gave 72 diagrams, 18 site-track records and 54 reference-alternate pairs. Selection preceded the Julia outcomes but deliberately included known exceptions; this was targeted computational validation, not a new prospective biological sample.

Each map retained all 512 bins and 130,816 unordered edges. The Julia runner received raw maps and identifiers, not expected diagrams, and independently reconstructed the complete weighted flag filtration over $\mathbb{F}_2$. It reproduced the original offset and effective float32 edge inputs, used an infinite filtration threshold with no positive-lifetime censoring, and restored the original float64 offset after converting output coordinates to float64. Comparison required exact equality of the complete interval multisets, preserving repeated intervals rather than deduplicating them. All 72 diagrams, comprising 847,288 intervals, matched exactly. Effective input hashes, offsets and independently computed map means also agreed.

Before the genomic calculation, 22 small-graph controls matched a separate explicit boundary-matrix reduction over $\mathbb{F}_2$: ten named controls with hand-derived answers and twelve fixed-seed random graphs. Controls covered negative and zero births, ties, repeated intervals, nonmetric weights, an inactive edge, float32 near-tie collapse, translation and ignored matrix diagonals. Four deliberately corrupted comparisons were rejected: a missing duplicate interval, a changed endpoint, an unrestored offset and a corrupted input hash.

Downstream reconstruction from the Julia diagrams matched all 108 native/centered bottleneck distances and all 54 shifted Hausdorff witnesses exactly; all 2,565 recorded checks passed. The witness calculation used a separate blockwise all-pairs implementation with diagonal alternatives. Top sets and fractional credits matched at tolerances 0, 10^{-6} and 10^{-5}; stored GM12878 bound predictions were checked at 10^{-6} and IMR-90 predictions at all three tolerances. All three known IMR-90 exceptions remained: site 48/ALT index 2, site 51/ALT index 1 and site 59/ALT index 0. Their absolute saturation residuals were respectively 0.001204502435174145, 0.002583775381601489 and 0.00014162732143801343. Neither the full-cohort results nor the scientific conclusions changed.

Ripserer is a separate implementation in the Ripser algorithm family. Downstream bottleneck matching retained the shared GUDHI/Hera solver at approximation parameter zero. Exact agreement concerns the effective float32 filtration inputs and restored coordinates; it does not certify unrounded real-valued

maps, the complete numerical error allowance, or every site in the study. The check does not establish independent-model transfer, biological validity or clinical superiority. It supplies no Lean formalization and does not repeat the separate exact-rational conditional label tests.

The protocol, Julia runner, genomic inputs and primary comparator remained unchanged during genomic execution. After the initial source freeze, three receipt identity/hash assertions and a corrected explanatory docstring were added to the control comparator; control inputs and expected results were unchanged, and the stricter comparison passed. The original files, amendment and execution receipts are retained. Pinned Julia and Python environments, raw selected inputs, computed diagrams, comparison code, outputs and logs are supplied in the separate Julia validation capsule, 88,338,909 bytes, SHA-256:

`e795faca8ba6217ccc4a2876f8a2613095f8dc8c175c86d41909092a7a36df54`

The capsule was published as an addendum validating manuscript v0.2.3; its preserved report describes that earlier manuscript state. It is separate from the v0.2.0 full-study archive. Portable workspace preparation and relocated controls were checked; those packaging checks are not a second complete genomic replay. The source and result records are also available in the versioned Julia validation directory. These checks supplement the earlier validation harness and author review; they are not external human peer review.

Cited resources

1. Iben P. Reproducibility package for "What do within-site comparisons of AlphaGenome Atlas predictions measure? An audit at ClinVar sites in 11 genes". Zenodo, version 1.0, 2026. doi:10.5281/zenodo.23537469.
2. Čufar M. Ripserer.jl: flexible and efficient persistent homology computation in Julia. *Journal of Open Source Software*. 2020;5(54):2614. doi:10.21105/joss.02614.

Fixed illustrative site

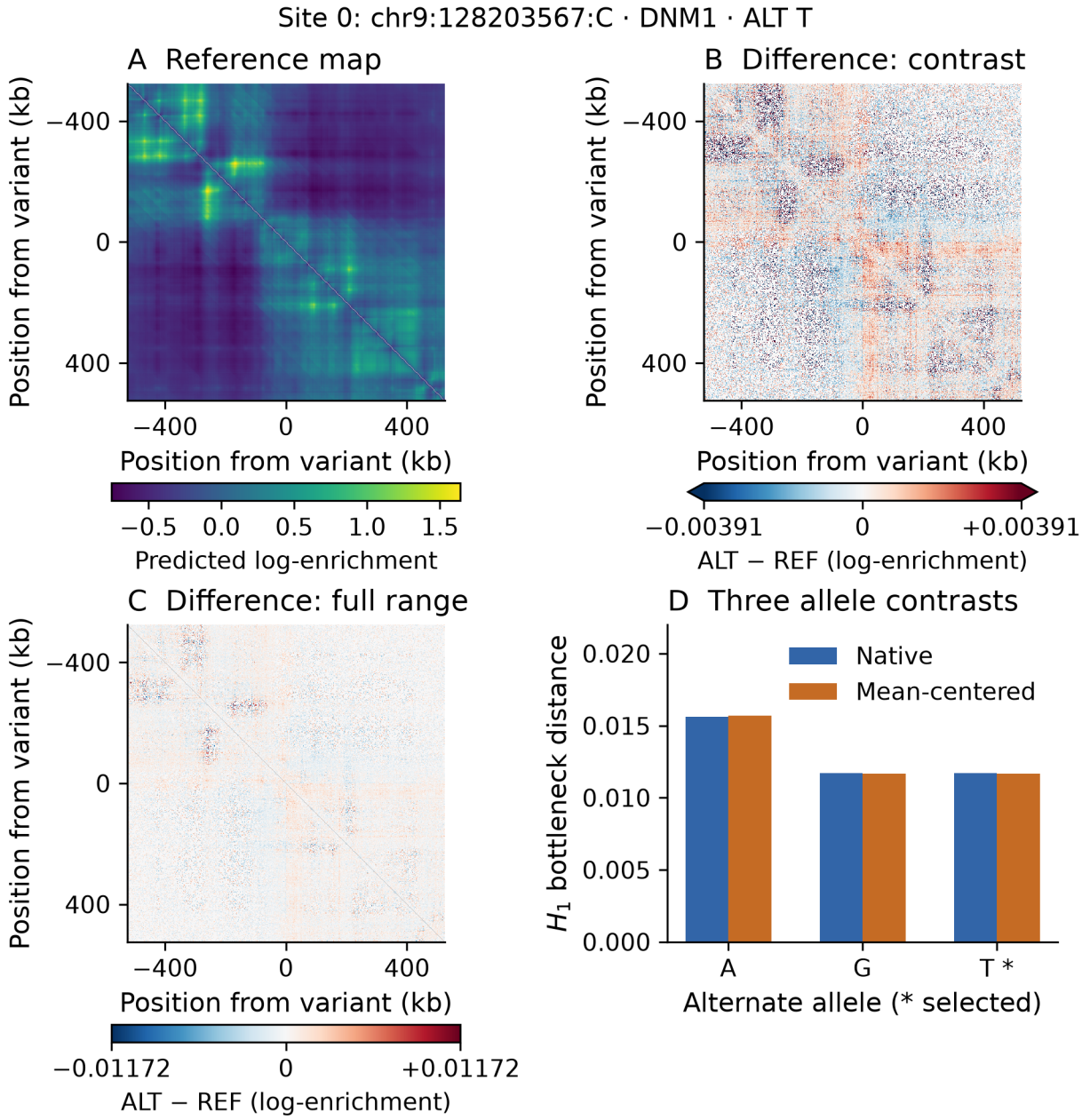


Figure S6.1. Fixed input-order site zero (chr9:128203567:C; DNM1; GM12878), not selected by effect size. **A**, reference predicted log-enrichment. **B**, selected-alternate T minus reference, with linear limits ±0.00390625 (99th percentile of absolute off-diagonal differences): 788/261632 cells (0.301%) color-clipped. **C**, identical differences at full linear range ±0.01171875, without clipping. Both scales are zero-centered in log-enrichment difference units. **D**, native and mean-centered H1 bottleneck distances for three alternate alleles; the asterisk marks the ClinVar-selected allele. Gray diagonals are excluded. No physical loop validation is implied. AlphaGenome-derived material remains subject to provider Output Terms identified in the main article.